

Class 10 NCERT Maths

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Chapter 6: Triangles

CHAPTER 6: TRIANGLES

A simplified, detailed guide in Hinglish

For students who missed Class 7-8-9

Based on the official NCERT English Medium textbook

SECTION 1: FOUNDATION

1.1 TRIANGLE KI BASICS

- Triangle = 3 lines, 3 angles
- Sum of angles = 180 degrees
- Types (by sides):
 - Equilateral: sab sides equal, sab angles 60
 - Isosceles: 2 sides equal
 - Scalene: sab sides different
- Types (by angles):
 - Acute: sab angles < 90
 - Right: ek angle = 90
 - Obtuse: ek angle > 90

1.2 CONGRUENT VS SIMILAR (CRITICAL!)

Property	Congruent	Similar
Shape	Same	Same
Size	SAME	MAY DIFFER
Sides	Equal	PROPORTIONAL
Angles	Equal	Equal

EASY:

Photo aur uski xerox copy = congruent

Photo aur uski enlarged = similar

1.3 SIMILAR FIGURES

Two figures similar agar:

1. Corresponding ANGLES equal ho
2. Corresponding SIDES proportional ho

EXAMPLES:

- Sab squares similar (lekin sab congruent nahi)
- Sab equilateral triangles similar
- Sab circles similar

SECTION 2: SIMILAR TRIANGLES

2.1 DEFINITION

Triangle ABC aur Triangle PQR similar agar:

1. angle A = angle P
angle B = angle Q
angle C = angle R
2. $AB/PQ = BC/QR = CA/RP$

Likha jaata hai: Triangle ABC ~ Triangle PQR

IMPORTANT: Letters ka ORDER matters - corresponding angles dikhata hai.

2.2 EXAMPLE

Triangle ABC ~ Triangle DEF
AB=6, BC=8, CA=10, DE=9.
Find EF and FD.

$$\begin{aligned} AB/DE &= BC/EF = CA/FD \\ 6/9 &= 8/EF = 10/FD \\ 2/3 &= 8/EF \rightarrow EF = 12 \\ 2/3 &= 10/FD \rightarrow FD = 15 \end{aligned}$$

So EF = 12, FD = 15.

SECTION 3: BPT (BASIC PROPORTIONALITY THEOREM)

3.1 STATEMENT (AKA THALES THEOREM)

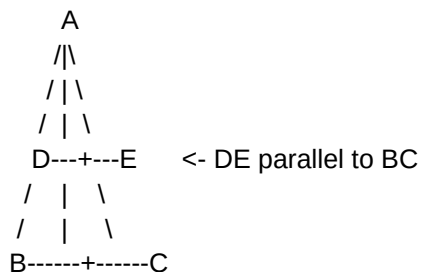
"If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, then the other two sides are divided in the same ratio."

AASAAN BHASHA:

Triangle ke ek side ke parallel line khichi jaye,

jo baaki dono sides ko cut kare, toh wo dono sides
SAME RATIO mein divide hote hain.

3.2 DIAGRAM



Agar DE parallel BC, toh:

$$AD/DB = AE/EC$$

3.3 CONVERSE OF BPT

"If a line divides any two sides of a triangle in the same ratio, then the line is parallel to the third side."

Agar $AD/DB = AE/EC$, toh DE parallel BC.

3.4 EXAMPLE

Triangle ABC mein DE parallel BC.
AD = 2.5 cm, DB = 3 cm, AE = 4 cm. EC = ?

$$\begin{aligned}AD/DB &= AE/EC \\2.5/3 &= 4/EC \\EC &= (4 \times 3)/2.5 = 4.8 \text{ cm}\end{aligned}$$

SECTION 4: SIMILARITY CRITERIA

3 main criteria hain - ek bhi satisfy ho toh
triangles similar hain.

4.1 AAA (ANGLE-ANGLE-ANGLE) SIMILARITY

Do triangles ke corresponding TEEN angles equal
ho - similar hain.

4.2 AA (ANGLE-ANGLE) SIMILARITY

Do triangles ke corresponding DO angles equal
ho - similar hain.

(Kyunki sum = 180, do same hone se teesra bhi same)

MOST USED CRITERION!

4.3 SSS (SIDE-SIDE-SIDE) SIMILARITY

Do triangles ke corresponding TEENO sides
PROPORTIONAL ho - similar hain.

4.4 SAS (SIDE-ANGLE-SIDE) SIMILARITY

Do triangles ke DO sides proportional aur unke
BEECH KA angle equal ho - similar hain.

4.5 EXAMPLE

Triangle ABC: AB=4, BC=6, AC=8
Triangle DEF: DE=6, EF=9, DF=12

Check ratios:

$$AB/DE = 4/6 = 2/3$$

$$BC/EF = 6/9 = 2/3$$

$$AC/DF = 8/12 = 2/3$$

Sab same -> Triangle ABC ~ Triangle DEF (by SSS)

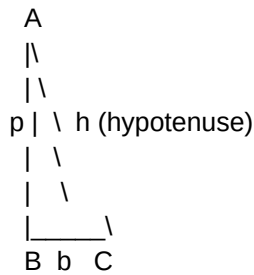
SECTION 5: PYTHAGORAS THEOREM

5.1 STATEMENT (FAMOUS!)

"In a right-angled triangle, the SQUARE of the
hypotenuse is equal to the SUM of squares of the

other two sides."

DIAGRAM:



If angle B = 90:

$$AC^2 = AB^2 + BC^2$$

$$h^2 = p^2 + b^2$$

5.2 PYTHAGOREAN TRIPLETS (YAAD RAKH)

Famous triplets - exam mein bahut use hote hain:

$$3, 4, 5 \quad \rightarrow \quad 9 + 16 = 25$$

$$5, 12, 13 \quad \rightarrow \quad 25 + 144 = 169$$

$$8, 15, 17 \quad \rightarrow \quad 64 + 225 = 289$$

$$7, 24, 25 \quad \rightarrow \quad 49 + 576 = 625$$

5.3 CONVERSE OF PYTHAGORAS

"Agar triangle ke kisi side ka square baaki do sides ke squares ke sum ke barabar ho, toh wo triangle right-angled hota hai."

5.4 EXAMPLE

3, 4, 5 sides wala triangle right-angled hai?

Largest side = 5

$$5^2 = 25$$

$$3^2 + 4^2 = 9 + 16 = 25 \quad (\text{same})$$

Haan, right-angled triangle hai!

Q AND A TIME

Q1. Congruent aur similar mein basic difference?

Q2. Triangle PQR ~ Triangle XYZ. PQ=6, QR=8, RP=10, XY=9. YZ aur ZX nikaalo.

Q3. Triangle ABC mein DE parallel BC. AD=1.5, DB=3, AE=1. EC = ?

Q4. Konsi similarity criterion lagegi?

- (a) Do triangles ke 2 angles equal
- (b) Do triangles ke saare 3 sides proportional
- (c) Do triangles ke 2 sides proportional aur included angle equal

Q5. Pythagoras: 6 cm aur 8 cm sides wale right triangle ka hypotenuse?

Q6. Check: kya 5, 12, 13 right triangle ke sides hain?

SUMMARY

1. Similar triangles: angles equal + sides proportional.

2. Congruent vs Similar:

Congruent = same size, same shape

Similar = same shape, different size

3. BPT (Thales): Parallel line in triangle splits other two sides in same ratio.

DE parallel BC $\rightarrow$ $AD/DB = AE/EC$

4. Similarity Criteria:

AAA / AA: angles equal

SSS: all sides proportional

SAS: 2 sides proportional + included angle

5. Pythagoras: $hyp^2 = side1^2 + side2^2$

6. Famous triplets: (3,4,5), (5,12,13), (8,15,17),

(7,24,25)
